Economy Topic 8 — Employment, Poverty, Human Capital and Social Economy

Comprehensive Study & Revision Notes

> **Key Reference Framework:** NCERT Class 11 Indian Economic Development (Poverty, Human Capital), UNDP Human Development Reports, NITI Aayog National MPI, MoSPI Periodic Labour Force Survey (PLFS), Census of India, and State Public Service Commission Examination archives.
> **Exam Weightage:** Very High — Repeatedly tested on poverty lines (Alagh, Tendulkar, Rangarajan), MPI dimensions, HDI formulation, demographic dividend age group (15–59), and core national welfare flagship schemes.

Current Affairs & Key Indicators

Year / Index	Key Development & Metric	Core Analytical & Policy Significance
Human Development Report (HDR 2025)	India ranked **130th** (HDI value ≈ 0.644); life expectancy at birth reached **72.0 years** (up from 58.6 years in 1990).	Reflects long-term gains in child survival and longevity; knowledge and income components remain areas for policy intervention.
World Happiness Report 2026	India ranked approximately **116th** globally in subjective well-being.	Evaluates subjective well-being and social support; distinct from the composite capability framework of the HDI.
Multidimensional Poverty Index (MPI)	Introduced in **2010** by UNDP and OPHI, replacing the Human Poverty Index (HPI, 1997).	Measures acute non-monetary deprivations across health, education, and living standards; does not use income poverty lines.
Poverty Social Indicators	Core indicators encompass illiteracy, lack of drinking water, sanitation, and joblessness.	Evaluates capability deprivations; dwelling floor area or size of house is not treated as a social indicator of poverty.
Inaugural HDR (1990)	Led by Pakistani economist **Mahbub ul Haq** with conceptual support from **Amartya Sen**.	Shifted the development paradigm from national income accounting to people-centric human capability expansion.
Human Capital Returns	Public and private investment in education and healthcare enhances human capital.	Directly elevates labor productivity, technological adoption, and lifetime earning capacity.
HDI Dimensions	Computed from three core dimensions: **health, education, and standard of living**.	Measures life expectancy, schooling years, and GNI per capita; environmental quality is not an included indicator.
Demographic Dividend	Defined by a high share of working-age population (**15–59 years** in Indian planning convention).	Represents a transitory window of economic growth potential requiring investments in health, skills, and employment.

Year / Index	Key Development & Metric	Core Analytical & Policy Significance
Historical Poverty Line	**Dadabhai Naoroji** formulated the earliest poverty line using the **jail cost of living**.	Estimated subsistence consumption requirements in pre-Independence India (*Poverty and Un-British Rule in India*).

8.1 Poverty — Concepts, Poverty Lines, and Multidimensional Poverty

Poverty represents the inability of an individual or family to command sufficient economic resources to satisfy basic physical, nutritional, and social needs.

- **Absolute Poverty:** A condition characterized by severe deprivation of basic human needs, measured against a fixed, real consumption or income threshold that does not change with rising living standards of society.
- **Relative Poverty:** A condition where individuals experience economic disadvantage relative to the average living standard of their contemporary society, reflecting economic inequality and social exclusion.
- **Primary Methodology:** Official poverty measurement in India is based on **household consumer expenditure surveys (CES)** conducted by the National Sample Survey Office (NSSO, now NSO), rather than direct income reporting.
- **Head Count Ratio (HCR):** The proportion of the population falling below the officially designated poverty threshold.
- **Poverty Gap Index:** Measures the depth of poverty by calculating the average shortfall of the poor population below the poverty line, expressed as a percentage of the poverty line itself.

Key Committees on Poverty Estimation

- **Dadabhai Naoroji (Pre-Independence):** In his treatise *Poverty and Un-British Rule in India*, Naoroji computed the earliest subsistence poverty line for India based on the **jail cost of living** for prisoners, adjusting prices for children and non-working dependents.
- **National Planning Committee (1938):** Chaired by Jawaharlal Nehru, it established a subsistence living standard ranging from ₹15 to ₹25 per capita per month at pre-war prices.
- **Dandekar and Rath (1971):** Formulated the first systematic calorie-based poverty estimation using NSS data (1960–61), establishing a daily intake benchmark of **2,250 calories per person** across both rural and urban areas.
- **Alagh Task Force (1979):** Chaired by Y.K. Alagh, this Planning Commission task force recommended an official poverty line based exclusively on nutritional requirements: **2,400 kcal per day for rural areas** and **2,100 kcal per day for urban areas**.
- **Lakdawala Committee (1993):** Chaired by D.T. Lakdawala, it retained the Alagh nutritional calorie benchmarks but recommended calculating State-specific poverty lines updated through the Consumer Price Index for Agricultural Labourers (CPI-AL) for rural areas and the Consumer Price Index for Industrial Workers (CPI-IW) for urban areas.
- **Tendulkar Committee (2009):** Chaired by Suresh Tendulkar, it made major methodological revisions:

- Departed from anchoring poverty exclusively to calorie norms and adopted an expanded consumption basket that included private expenditures on health, education, clothing, and footwear.
- Formulated a uniform poverty line basket across all States anchored to the urban consumption basket.
- Utilized Mixed Reference Period (MRP) expenditure data from the NSS 61st Round (2004–05).
- For 2011–12, the Tendulkar poverty lines were determined at **₹816 per capita per month in rural areas** (approx. ₹26 per day) and **₹1,000 per capita per month in urban areas** (approx. ₹32 per day).
- Estimated India's overall poverty head count ratio at **21.9%** (25.7% rural, 13.7% urban) for 2011–12.
- **Rangarajan Committee (2014):** Chaired by C. Rangarajan, it recommended returning to a nutritionally grounded consumption basket complemented by non-food expenditures (clothing, housing, education, and conveyance).
- Proposed daily per capita poverty thresholds of **₹32 in rural areas** (₹972/month) and **₹47 in urban areas** (₹1,407/month) at 2011–12 prices.
- Estimated the all-India poverty ratio at **29.5%** for 2011–12; the report was submitted to the Planning Commission but was not adopted as the live official series.

Inequality and Structural Theories of Poverty

- **Lorenz Curve:** A graphical representation of the cumulative distribution of national income or wealth against the cumulative share of the population. The degree of deviation from the 45-degree line of perfect equality illustrates the extent of inequality.
- **Gini Coefficient:** A mathematical index derived from the Lorenz curve measuring the ratio of the area between the 45-degree line of equality and the Lorenz curve to the total area beneath the equality line. It ranges from 0 (perfect equality) to 1 (maximal inequality).
- **Vicious Circle of Poverty:** Formulated by economist **Ragnar Nurkse**, demonstrating that poverty is self-perpetuating on both supply and demand sides: low real income causes low savings and low purchasing power, which leads to low capital investment, generating low productivity, and culminating back in low income.
- **Culture of Poverty:** Propounded by anthropologist **Oscar Lewis**, positing that prolonged poverty generates a distinct set of shared cultural norms, fatalistic attitudes, and survival behaviors transmitted across generations.
- **Cyclical Poverty:** Refers to transient poverty where households fluctuate above and below the poverty line depending on macroeconomic business cycles or seasonal agricultural shocks.

Social Indicators and Deprivation Assessment

- Comprehensive poverty evaluations examine broader socio-economic capability deprivations:
- Persistent adult and female illiteracy.
- Child malnutrition, stunting, and wasting.
- Lack of access to clean and treated drinking water.
- Inadequate sanitation and open defecation practices.
- Chronic lack of gainful, secure employment opportunities.
- **Housing Quality vs Size:** Social deprivation metrics assess structural housing security (e.g., kuccha vs pucca structures, floor and roof materials), but do not consider dwelling floor area or "size of

house" as an indicator of poverty.

Multidimensional Poverty Index (MPI)

- **Genesis:** Developed in 2010 by the Oxford Poverty and Human Development Initiative (OPHI) and the United Nations Development Programme (UNDP), formally replacing the earlier **Human Poverty Index (HPI)** introduced in 1997.
- **Non-Monetary Framework:** The MPI evaluates deprivations experienced simultaneously at the household level across ten indicators grouped into three equally weighted dimensions: 1. **Health (Weight 1/3):** Nutrition, Child Mortality. 2. **Education (Weight 1/3):** Years of Schooling, School Attendance. 3. **Standard of Living (Weight 1/3):** Cooking Fuel, Sanitation, Drinking Water, Electricity, Housing, Assets.
- **Deprivation Cut-off:** A household is categorized as multidimensionally poor if it is deprived in 33.3% or more of the weighted indicators.
- **National MPI (NITI Aayog):** NITI Aayog developed India's official National Multidimensional Poverty Index based on NFHS microdata, expanding the global 10 indicators to 12 by incorporating **Maternal Health (Antenatal Care)** and **Bank Accounts**.

Q-EC1. UPPCS (Pre) 2020

Which of the following Committees recommended a poverty line based on nutritional requirements exclusively?

A. Alagh Committee B. Lakdawala Committee C. Tendulkar Committee D. Rangarajan Committee

Show answer

Explanation: The Task Force on Projections of Minimum Needs and Effective Consumption Demand (1979), chaired by Y.K. Alagh, established India's official poverty line anchored exclusively to nutritional requirements: 2,400 kcal per person per day for rural areas and 2,100 kcal per person per day for urban areas. Later committees (Tendulkar and Rangarajan) broadened the basket to incorporate expenditures on health, education, clothing, and shelter.

Ans: A. Alagh Committee

Q-EC2. UPPCS (Pre) 2020

Human Development Report 1997 introduced the concept of Human Poverty Index (HPI) but Human Development Report replaced it with Multidimensional Poverty Index (MPI) in the year

A. 1999 B. 2005 C. 2010 D. 2015

Show answer

Explanation: The United Nations Development Programme (UNDP) published the Human Poverty Index (HPI) beginning in 1997, but formally replaced it with the Multidimensional Poverty Index (MPI) in the 2010 Human Development Report, developed in collaboration with the Oxford Poverty and Human Development Initiative (OPHI).

Ans: C. 2010

Q-EC3. UPPCS (Pre) 2025

With reference to the Multidimensional Poverty Index (MPI), which of the following statements is/are correct?

1. It was developed and introduced in 2010.
2. It measures extreme poverty and includes people living on less than one dollar per day.

Select the correct answer from the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

Show answer

Explanation: Statement 1 is correct: The global MPI was developed by UNDP and OPHI and introduced in the 2010 Human Development Report. Statement 2 is incorrect: The MPI is a non-monetary index measuring acute overlapping deprivations across health, education, and living standards; the income metric of living on less than USD 1/day (now USD 2.15/day) is the World Bank's international monetary extreme poverty line.

Ans: D. Only 1

Q-EC4. UPPCS (Pre) 2019

Multidimensional poverty index includes:

i. Health ii. Education iii. Living Standard

Select correct answer from the codes given below:

A. Only i is correct B. Only ii and iii are correct C. Only i and ii are correct D. i, ii and iii are correct

Show answer

Explanation: The Multidimensional Poverty Index (MPI) is structured around three equally weighted dimensions mirroring the Human Development Index: Health, Education, and Living Standard, assessed through ten specific indicators.

Ans: D. i, ii and iii are correct

Q-EC5. UPPCS (Pre) 2025

Which of the following is NOT a social indicator of poverty?

1. Illiteracy
2. Safe drinking water
3. Employment opportunities
4. Size of house

Select the correct answer from the code given below:

A. 1 and 2 B. 3 and 4 C. Only 4 D. Only 1

Show answer

Explanation: Standard social indicators of poverty assess social capabilities and deprivations such as illiteracy, lack of access to safe drinking water, malnutrition, and lack of employment opportunities. While housing quality (materials used for roof, walls, and flooring) is assessed in living standards, dwelling floor area or "size of house" is not considered a social indicator of poverty.

Ans: C. Only 4

Q-EC6. UKPCS (Pre) 2025

Who used the 'jail cost of living' to estimate the poverty line in India?

A. Dadabhai Naoroji B. Mahatma Gandhi C. C. D. Deshmukh D. Vallabh Bhai Patel

Show answer

Explanation: In his seminal work *Poverty and Un-British Rule in India*, Dadabhai Naoroji computed the earliest subsistence poverty line for colonial India by calculating the "jail cost of living" (the cost of food and clothing necessary to keep an adult prisoner alive and functional) and making mathematical adjustments for children and women.

Ans: A. Dadabhai Naoroji

8.2 Employment, Labour Force Metrics, and Demographic Dividend

- **Unemployment:** Defined as a condition wherein individuals belonging to the working-age group, who are physically capable and actively seeking work at the going wage rate, cannot secure gainful employment.

Classification of Unemployment

- **Disguised Unemployment:** A structural phenomenon in which more labor is employed in an activity than is technically required, such that the marginal productivity of additional workers is zero or negative. Prevalent extensively in Indian peasant agriculture.
- **Seasonal Unemployment:** Occurs when labor is rendered idle during specific seasonal intervals, widespread in agriculture, sugar milling, and tourism.

- **Structural Unemployment:** Caused by fundamental long-term shifts in economic structure, technological obsolescence, or educational deficits creating an imbalance between market skill requirements and job seekers' qualifications.
- **Frictional Unemployment:** Temporary joblessness experienced when workers are between jobs or re-entering the labor force to find better employment matches.
- **Cyclical Unemployment:** Induced by deficient aggregate demand during macroeconomic recessions or economic slowdowns.

Indian Labour Force Concepts (PLFS Framework)

- The National Sample Survey Office (NSO) measures labor force activity under three standard approaches: 1. **Usual Principal and Subsidiary Status (UPSS):** Assesses activity status over a 365-day reference period; captures chronic and long-term employment. 2. **Current Weekly Status (CWS):** Assesses activity over a 7-day reference period; an individual is considered employed if working for at least one hour on any day during the reference week. 3. **Current Daily Status (CDS):** Measures employment day-by-day (in half-day units), capturing short-term underemployment.
- **Labour Force:** The sum of all employed persons plus all unemployed persons actively seeking or available for employment.
- **Labour Force Participation Rate (LFPR):** The percentage of the working-age population that is part of the labour force:

> **Formula:** `LFPR = (Labour Force / Total Working-Age Population) × 100`

- **Worker Population Ratio (WPR):** The percentage of employed persons relative to the total population.
- **Unemployment Rate (UR):** The percentage of unemployed persons relative to the total labour force:

> **Formula:** `UR = (Unemployed Persons / Labour Force) × 100`

- **Informal and Gig Employment:** The Indian economy features a high proportion of informal employment characterized by minimal social protection. Digital platform and gig work have introduced flexible labor arrangements, prompting social security frameworks under the Code on Social Security, 2020.

The Demographic Dividend

- **Conceptual Definition:** The transitory economic growth surge that occurs when a nation's dependency ratio drops due to a high proportion of the population falling into the productive working-age category relative to young and aged dependents.
- **Working-Age Cohort:** In Indian demographic planning and official statistical conventions, the productive working-age population is classified as **15 to 59 years** (as compared to 15 to 64 years in broader UN classifications).
- **Policy Imperative:** Realizing the demographic dividend requires substantial public investment in skill development, quality education, preventive healthcare, and industrial job creation; without these investments, demographic pressure risks manifesting as demographic liability.

Q-EC7. UPPCS (Pre) 2022

The demographic dividend is—

A. Working population of above 60 years B. Working population of 15 to 59 years C. 0–6 years population D. Working population of 14 to 50 years

Show answer

Explanation: The demographic dividend arises from a favorable age structure where the working-age population exceeds the dependent population. In Indian demographic classification and planning literature, the productive working-age cohort is officially categorized as 15 to 59 years.

Ans: B. Working population of 15 to 59 years

8.3 Human Capital Formation, Human Development Index, and Inclusive Growth

- **Human Capital:** The acquired knowledge, skills, technical competencies, and physical vitality embodied in the working population.
- **Determinants of Human Capital:** Investment in primary and higher education, vocational skill training, preventive and curative public health, labor market information systems, and migration for employment opportunities.
- **Economic Returns:** Education and healthcare enhance individual cognitive abilities, productivity, and resilience, generating higher lifetime earnings and driving aggregate economic output.

Human Development Index (HDI)

- **Origins:** Inaugurated in 1990 by the United Nations Development Programme (UNDP) under the leadership of Pakistani economist **Mahbub ul Haq**, intellectually guided by **Amartya Sen's Capability Approach**.
- **Three Core Dimensions and Indicators:** 1. **Long and Healthy Life:** Measured by **Life Expectancy at Birth** (fixed goalposts: minimum 20 years, maximum 85 years). 2. **Knowledge:** Measured as the arithmetic mean of:
  - **Mean Years of Schooling** for adults aged 25 and older.
  - **Expected Years of Schooling** for school-age children. 3. **Decent Standard of Living:** Measured by **Gross National Income (GNI) per capita (PPP USD)**. (Note: Post-2010 replaced GDP per capita with GNI per capita to reflect net primary cross-border income).
- **Aggregation Method:** Computed as the **geometric mean** of the three normalized dimension indices:

  > **Formula:** `HDI = (I_Health × I_Education × I_Income)^(1/3)`

- **Distinguishing Boundaries:** Environmental quality, political freedoms, and subjective well-being are excluded from the HDI calculation.

Alternative Welfare Measures and Indices

- **Physical Quality of Life Index (PQLI):** Formulated in 1979 by **Morris D. Morris** for the Overseas Development Council, averaging three non-income indicators: Infant Mortality Rate, Life Expectancy at Age 1, and Basic Literacy Rate.
- **Historical Chronology of Progress Metrics:**

> **Chronological Evolution:** `Per Capita Income (PCI) → Physical Quality of Life Index (PQLI) → Human Development Index (HDI)`

- **Capability Approach:** Developed by Nobel laureate **Amartya Sen**, positing that economic development should be evaluated through the expansion of individuals' substantive capabilities (what people are actually able to do and be) rather than utility or commodity possession alone.
- **Gender Inequality Index (GII):** Introduced in HDR 2010 to measure gender-based disparities across three dimensions: reproductive health (maternal mortality ratio and adolescent birth rate), empowerment (share of parliamentary seats and attainment of secondary education), and labor market participation.
- **Gross National Happiness (GNH):** Coined in 1972 by Bhutan's Fourth King, Jigme Singye Wangchuck, measuring national progress through sustainable socio-economic development, cultural preservation, environmental protection, and good governance.
- **World Happiness Report:** Published by the UN Sustainable Development Solutions Network (SDSN), ranking nations based on subjective evaluations of life satisfaction from the Gallup World Poll.
- **State HDRs in India: Madhya Pradesh** became the first State in India to release a State Human Development Report in **1995**. In national comparative rankings, **Kerala** consistently achieves the highest HDI score, whereas **Bihar** and **Uttar Pradesh** rank in lower quartiles.
- **Bare Necessities Index (BNI):** Designed in the Economic Survey 2020–21, tracking progress in basic necessities across **26 indicators** organized into **five dimensions** (water, sanitation, housing, micro-environment, and other facilities) using 2012 and 2018 data.

Sustainable Development Principles

- **Brundtland Report (*Our Common Future*, 1987):** Established the global benchmark definition: *"Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs."*
- ***The Limits to Growth* (1972):** Authored by Donella Meadows and team for the Club of Rome, demonstrating using computer simulations that continued exponential population and industrial growth would encounter finite ecological resource boundaries within 100 years.
- **Environmental Kuznets Curve (EKC):** An inverted-U-shaped hypothesis postulating that environmental degradation intensifies in early stages of economic development but stabilizes and declines once income passes a critical per capita threshold.
- **Carrying Capacity:** The ecological threshold indicating the maximum population and resource extraction that an ecosystem can sustainably support without undergoing irreversible degradation.

Q-EC8. UPPCS (Pre) 2025

Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Investment in human capital yields returns in the future.

Reason (R): Education and health make people more productive.

Select the correct answer from the code given below:

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Explanation: Human capital refers to the acquired knowledge, skills, technical competencies, and physical health embodied in a workforce. Public and private expenditures on education and healthcare enhance individuals' cognitive faculties and productivity, which in turn generate compounding economic returns in the future. Therefore, both statements are correct, and Reason (R) correctly explains Assertion (A).

Ans: D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Q-EC9. UPPCS (Pre) 2024

Which indicators are used in Human Development Index (HDI)?

1. Income level
2. Education
3. Condition of environment
4. Life expectancy

Select the correct answer from the codes given below:

A. Only 1, 2 and 3 B. Only 1 and 4 C. Only 1, 2 and 4 D. Only 1 and 2

Show answer

Explanation: The Human Development Index (HDI) is computed from three core dimensions: a long and healthy life (life expectancy at birth), knowledge (mean and expected years of schooling), and a decent standard of living (Gross National Income per capita (PPP USD)). Environmental conditions and sustainability are not included in the HDI formula.

Ans: C. Only 1, 2 and 4

Q-EC10. UPPCS (Pre) 2025

With reference to Human Development Index, which of the following statements is/are correct?

1. The UNDP published its first Human Development Report in 1990, which had a Human Development Index.
2. The first team that developed the Human Development Index was led by Mahbub-ul-Haq and S. Priesner.

Select the correct answer using the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

Show answer

Explanation: Statement 1 is correct: The United Nations Development Programme (UNDP) released the first Human Development Report featuring the HDI in 1990. Statement 2 is incorrect: The team was conceived and led by Pakistani economist Mahbub ul Haq with major conceptual input from Amartya Sen, Keith Griffin, Sudhir Anand, and Meghnad Desai; S. Priesner was not a team leader of the 1990 report.

Ans: D. Only 1

8.4 Major National Welfare, Employment, and Social Security Schemes

Transformational Planning Frameworks

- **Aspirational Districts Programme (ADP):** Launched by **NITI Aayog** in **January 2018** to rapidly transform **112** of the most underdeveloped districts in India, tracked through **49 Key Performance Indicators** across five focus areas: Health & Nutrition, Education, Agriculture & Water, Financial Inclusion & Skill, and Basic Infrastructure.
- **Aspirational Blocks Programme (ABP):** Launched in **January 2023** covering **500 underdeveloped blocks** to accelerate grassroots saturation of welfare deliverables.

Comprehensive Healthcare and Nutrition Missions

- **Ayushman Bharat – PM-JAY:** Launched on **23 September 2018** at **Ranchi (Jharkhand)**; provides comprehensive health insurance of **₹5 lakh per eligible family per year** for secondary and tertiary care hospitalization across public and empanelled private hospitals, implemented by the **National Health Authority (NHA)**.
- **Ayushman Bharat Digital Mission (ABDM):** Launched in **September 2021** to create an interoperable digital health ecosystem through Ayushman Bharat Health Account (ABHA) IDs; citizen and provider participation is voluntary.
- **National Rural Health Mission (NRHM):** Launched on **12 April 2005** during the Tenth Five-Year Plan, integrating public healthcare delivery and introducing the community-based **Accredited Social Health Activist (ASHA)** cadre.
- **Mission Indradhanush:** Launched on **25 December 2014**; guarantees full immunization coverage for children up to 2 years and pregnant women against vaccine-preventable diseases.
- **POSHAN Abhiyaan (National Nutrition Mission):** Launched in **March 2018** from Jhunjhunu (Rajasthan); targets the reduction of stunting, undernutrition, anemia, and low birth weight; observed through annual **Rashtriya Poshan Maah** every **September**.
- **Integrated Child Development Services (ICDS):** Centrally sponsored scheme launched on **2 October 1975**, delivering six core services: supplementary nutrition, immunization, health check-ups,

referral services, pre-school non-formal education, and nutrition/health education to children (0–6 years) and mothers.

National Skill Development and Livelihood Missions

- **Pradhan Mantri Kaushal Vikas Yojana (PMKVY):** Flagship skill initiative of the **Ministry of Skill Development and Entrepreneurship (MSDE)**, executed by the National Skill Development Corporation (NSDC), providing short-term certified training aligned with the National Skills Qualifications Framework (NSQF).
- **Skill Development Initiatives:** Includes schemes such as **STRIVE** (Skills Strengthening for Industrial Value Enhancement) and **SANKALP** (Skills Acquisition and Knowledge Awareness for Livelihood Promotion); in contrast, **STRIDE** is an academic research scheme under the University Grants Commission (UGC).
- **Deendayal Antyodaya Yojana – NRLM (DAY-NRLM):**
- Historical lineage: Restructured from the **Integrated Rural Development Programme (IRDP, 1980)** into the **Swarnajayanti Gram Swarozgar Yojana (SGSY, 1999)**.
- Re-launched as the **National Rural Livelihoods Mission (NRLM)** in **June 2011**, becoming fully operational in April 2013, and formally renamed **DAY-NRLM** on **29 March 2016**.
- Operational core: Mobilizes rural poor women into Self-Help Groups (SHGs) and federations, promoting sustainable livelihood enhancements through microfinance and revolving community investment funds.
- **Swarna Jayanti Shahari Rozgar Yojana (SJSRY):** Launched on **1 December 1997** to address urban unemployment and underemployment through self-employment micro-enterprises and wage employment; succeeded by the National Urban Livelihoods Mission (DAY-NULM) in 2013.

Statutory Rural Wage Guarantee — MGNREGA

- **Legislative Framework:** Enacted by Parliament as the National Rural Employment Guarantee Act in **2005**, notified on 7 September 2005, and brought into force on **2 February 2006** across an initial 200 districts.
- **Statutory Name:** Formally renamed the **Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)** on **2 October 2009**.
- **Core Entitlement:** Provides a statutory legal entitlement of **100 days of guaranteed unskilled manual wage employment per financial year** to every rural household whose adult members volunteer for manual work.
- **Work Application Guarantee:** If employment is not provided within **15 days** of application, the State is legally mandated to pay an unemployment allowance.
- **Social Audit:** Mandates mandatory periodic social audits through Gram Sabhas to ensure operational transparency and prevent fund leakages.

Social Security, Old-Age Pensions, and Financial Inclusion

- **Pradhan Mantri Jan Dhan Yojana (PMJDY):** Launched on **28 August 2014**; ensures universal access to basic banking accounts, overdraft facilities, and embedded RuPay accident insurance coverage.

- **Atal Pension Yojana (APY):** Launched on **9 May 2015**; targeted primarily at unorganised sector workers aged 18 to 40; guarantees a fixed monthly pension of ₹1,000 to ₹5,000 after age 60; upon the subscriber's death, the pension continues uninterrupted to the surviving spouse.
- **Pradhan Mantri Shram Yogi Maan-dhan (PM-SYM):** Launched in **2019**; voluntary contributory pension scheme for unorganised workers with monthly income up to ₹15,000; provides a guaranteed minimum pension of **₹3,000 per month** after age 60, with a 50% family pension payable solely to the surviving spouse.
- **National Social Assistance Programme (NSAP):** Operational since **15 August 1995**; administers Central financial assistance for the elderly (IGNOAPS), widows (IGNWPS), and persons with severe disabilities (IGNDPS) living below the poverty line.
- **Employees' State Insurance (ESI) Act, 1948:** Provides comprehensive social security benefits (medical care, sickness, maternity, and disablement benefits) to organized and notified commercial establishments.

Basic Rural and Urban Infrastructure Missions

- **Bharat Nirman (2005):** A targeted time-bound business plan for rural infrastructure comprising six components: **irrigation, drinking water, electrification, rural roads, rural housing, and rural telephony**.
- **Pradhan Mantri Gram Sadak Yojana (PMGSY):** Launched on **25 December 2000** to provide all-weather road connectivity to unconnected rural habitations with a population of 500 or more in plains (250+ in hill, desert, and tribal areas).
- **Jal Jeevan Mission (JJM):** Announced in August 2019 under the Ministry of Jal Shakti to provide functional household tap connections (FHTC) providing 55 litres of potable water per capita per day to every rural household.
- **Swachh Bharat Mission (SBM):** Launched on **2 October 2014** to eliminate open defecation and achieve universal sanitation by 2 October 2019, commemorating Mahatma Gandhi's 150th birth anniversary. SBM-Gramin Phase II focuses on sustaining ODF status and managing solid/liquid waste.
- **Pradhan Mantri Awas Yojana (PMAY):** Launched in **June 2015** with the goal of "Housing for All", covering urban housing (PMAY-U) and rural housing (PMAY-G, revamped from the earlier Indira Awas Yojana).
- **SVAMITVA Scheme:** Launched in **2020** by the Ministry of Panchayati Raj; deploys drone technology and Continuously Operating Reference Stations (CORS) to survey inhabited rural land (Abadi) and issue legal property cards.

8.5 Demography — Transition Stages, Census Chronology, and Indian Demographic Trends

Demography involves the systematic study of human populations, analyzing size, composition, spatial distribution, and dynamics driven by births, deaths, and migration.

Classical Demographic Transition Model

The demographic transition model delineates the long-term historical evolution of population growth through distinct phases:

1. **Stage 1 (High Stationary):** High birth rates and high death rates; population growth is low and fluctuates with epidemics and famines.
2. **Stage 2 (Early Expanding):** High birth rates coupled with sharply declining death rates due to advances in medicine, food security, and sanitation; leads to a rapid population explosion.
3. **Stage 3 (Late Expanding):** Falling birth rates alongside low death rates; population growth continues but at a decelerating rate.
4. **Stage 4 (Low Stationary):** Low birth rates and low death rates; population stabilizes with low natural increase.
5. **Stage 5 (Declining):** Birth rate falls below replacement level (Total Fertility Rate < 2.1), leading to an aging and contracting population.

- **India's Demographic Phase:** India has transitioned through Stage 2 and is currently positioned in **Stage 3 (Late Expanding)**, characterized by declining fertility rates and slowing population growth.
- **Natural Population Growth:** Calculated strictly as:

> **Formula:** `Natural Growth Rate = Crude Birth Rate (CBR) - Crude Death Rate (CDR)` *(Note: Natural growth excludes the effect of net international or domestic migration).*

Malthusian Theory of Population

- **Thomas Robert Malthus** published his influential work *An Essay on the Principle of Population* in **1798**.
- **Key Proposition:** Population, when unchecked, tends to grow in **geometric progression** (1, 2, 4, 8, 16...), whereas food and agricultural production increases only in **arithmetic progression** (1, 2, 3, 4, 5...).
- **Checks on Population:**
- **Positive Checks:** Natural and violent correctives such as famines, wars, diseases, and epidemics that elevate the death rate.
- **Preventive Checks:** Voluntary human measures such as moral restraint, delayed marriage, and birth control to curb the birth rate.

Census Chronology in India

- **Initial Census Efforts:** The first non-synchronous population census was initiated under **Lord Mayo** in **1872**.
- **First Synchronous Census:** The first comprehensive, synchronous decennial Census of India was conducted under Viceroy **Lord Ripon** in **1881**.
- **The "Year of the Great Divide" (1921):** The Census of 1921 is historically designated as the **Demographic Great Divide**, being the only decade (1911–1921) that recorded **negative decadal population growth (–0.31%)**, largely due to the 1918 influenza pandemic and widespread famine. Since 1921, India's population has registered continuous, uninterrupted expansion.

- **Peak Growth Era:** The highest decadal growth rate in post-Independence India was recorded in **1961–1971 (24.80%)**.
- **Census 2011 Overview:**
- Total population: **121.08 crore** (approximately 17.5% of the total world population).
- Decadal growth rate (2001–2011): **17.70%** (a significant decline from 21.54% in 1991–2001).
- Population density: **382 persons per square kilometre** (up from 325 in 2001).
- Overall sex ratio: **943 females per 1,000 males**; Child sex ratio (0–6 years): **919**.
- Effective literacy rate (population aged 7 and above): **73.0%** (Male: 80.9%, Female: 64.6%).

State-Level Population Metrics (Census 2011)

- **Most Populous States:** Uttar Pradesh (19.98 crore) > Maharashtra (11.23 crore) > Bihar (10.40 crore) > West Bengal (9.12 crore).
- **Population Density Ranking: Bihar (1,106 persons/sq km)** is the densest State, followed by West Bengal (1,028), Kerala (860), and Uttar Pradesh (829). **Arunachal Pradesh (17 persons/sq km)** has the lowest density.
- **Decadal Growth Variations: Meghalaya (27.95%)** recorded the highest decadal growth rate among States, while **Nagaland (–0.58%)** was the only State to record negative population growth during 2001–2011.
- **Literacy Leaders: Kerala (94.0%)** ranks first in literacy, followed by Mizoram (91.3%); **Bihar (61.8%)** registered the lowest literacy rate.

8.6 Urbanization — Concepts, Census Criteria, and Urban Dynamics

Urbanization denotes the spatial reallocation of population from rural to urban areas, accompanied by structural shifts from agrarian employment to industrial and tertiary sectors.

Census of India Criteria for Urban Settlements

The Census classifies settlements into two statutory categories:

1. **Statutory Towns:** All places with a municipality, municipal corporation, cantonment board, or notified town area committee established by law.
2. **Census Towns:** Settlements that satisfy all three of the following demographic criteria simultaneously: * A minimum total population of **5,000 persons**. * At least **75% of the male main working population** engaged in non-agricultural pursuits. * A population density of at least **400 persons per square kilometre**. *(Note: Geographic territorial expansion or urban sprawl alone does not satisfy the criteria for a Census Town).*

Urbanization Trends in India (Census 2011)

- **National Urban Share:** India's urban population in 2011 stood at **31.16%** (approx. 37.7 crore persons), while the rural population accounted for **68.84%** (approx. 83.3 crore).
- **Pace of Urbanization:** India exhibits a moderate level of urbanization globally; however, in absolute numbers, India's urban population surpasses the total national population of the United States.
- **State Urbanization Rankings:**

- **Most Urbanized State by Proportion: Goa (62.17%)**, followed by Mizoram (52.11%) and Tamil Nadu (48.40% among larger States).
- **Least Urbanized State: Himachal Pradesh (10.03%)**, followed by Bihar (11.29%).
- **Largest Urban Population in Absolute Terms: Maharashtra (5.08 crore)**, followed by Uttar Pradesh (4.44 crore).
- **Million-Plus Urban Agglomerations:** The number of cities with a population of one million or more increased from **35 in 2001** to **53 in 2011**.
- **Top Urban Agglomerations:** Greater Mumbai, followed by Delhi and Kolkata.
- **Uttar Pradesh Million-Plus Cities (2011):** Seven cities: Kanpur, Lucknow, Ghaziabad, Agra, Varanasi, Meerut, and Allahabad (Prayagraj).

8.7 Global Demography and World Urbanization Dynamics

- **Global Population Milestones:** The world population crossed 5 billion in **1987**, 6 billion in **1999**, 7 billion in **2011**, and reached **8 billion on 15 November 2022**.
- **World Population Day:** Celebrated annually on **11 July**, commemorating "Five Billion Day" on 11 July 1987 to focus international attention on population issues and reproductive rights.
- **UN Population Publications:** The *State of World Population* report is issued annually by the **United Nations Population Fund (UNFPA)**.
- **Global Geographic Distribution:** Over 50% of the world's population resides in the temperate latitudinal zone between **20°N and 40°N**.
- **Continental Comparisons: Asia** accounts for approximately 60% of the global population and is the densest continent; **Africa** exhibits the fastest annual population growth rate.
- **Global Urbanization Levels:** Approximately **55%** of the world population lives in urban areas (projected by the UN to reach 68% by 2050). The most urbanized regions include North America (82%) and Latin America (81%), with Oceania/Australia also recording very high urbanization proportions.
